



University of Lagos

Department of Petroleum Engineering

**PGG 453
Subsea Engineering**

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Course Outline

- Overview of Subsea Engineering.
- Subsea Field Development
- Subsea Distribution System
- Subsea Surveying
- Positioning and Foundation
- Subsea installation and Vessels
- Subsea Cost Estimation
- Subsea Control
- Subsea Project Execution and Interfaces
- Subsea Risk and Reliability
- Subsea Equipment Risk Based inspection

Course Objective:

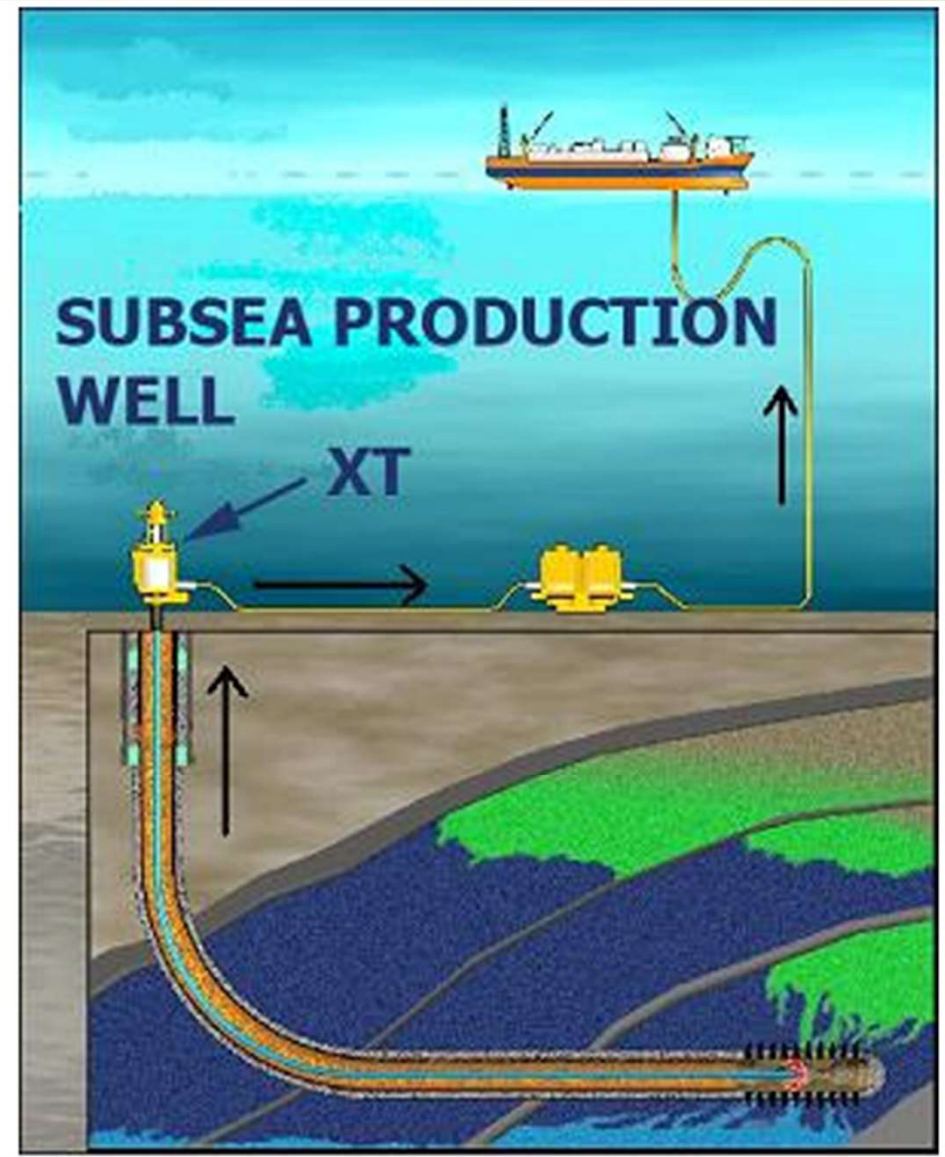
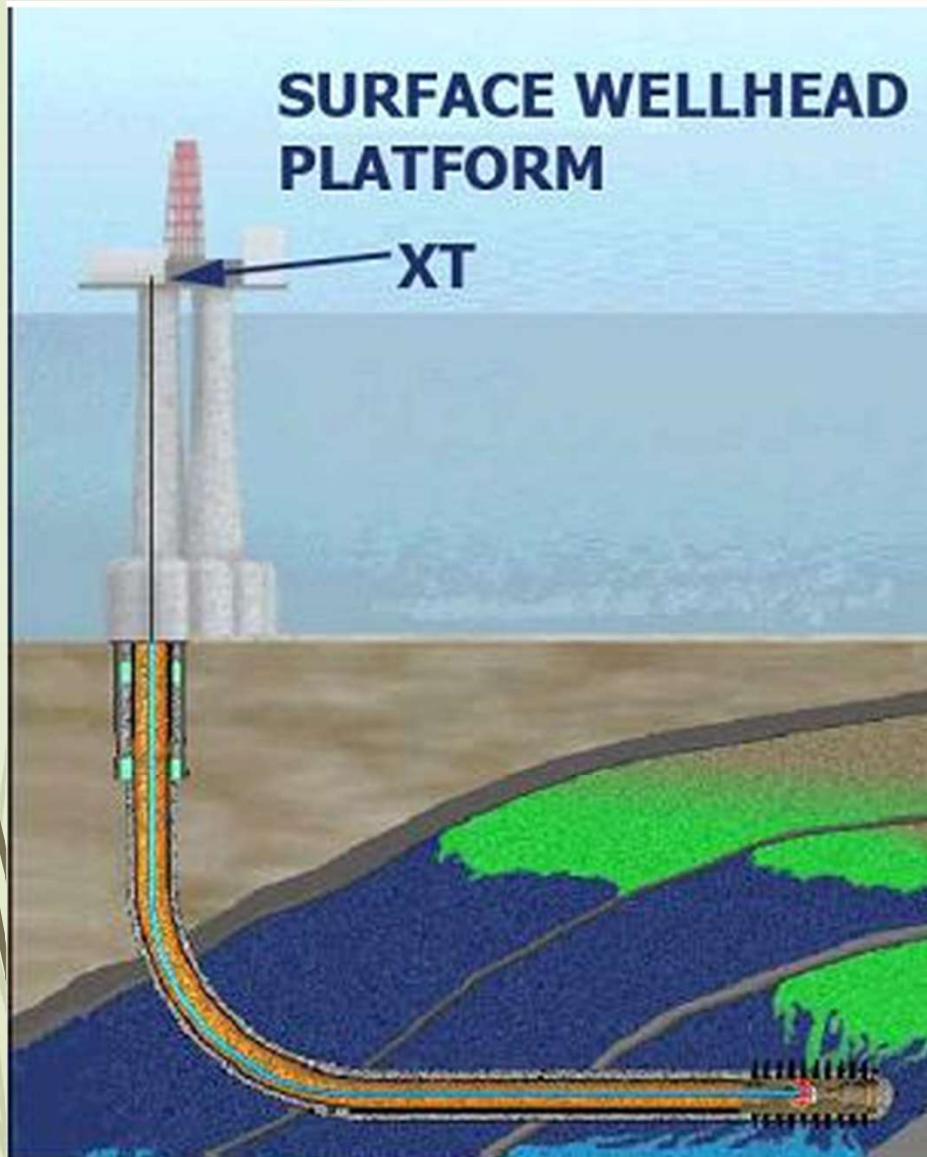
By the end of this course, students will:

Learn and understand Subsea Developments, Equipment, and Installations.

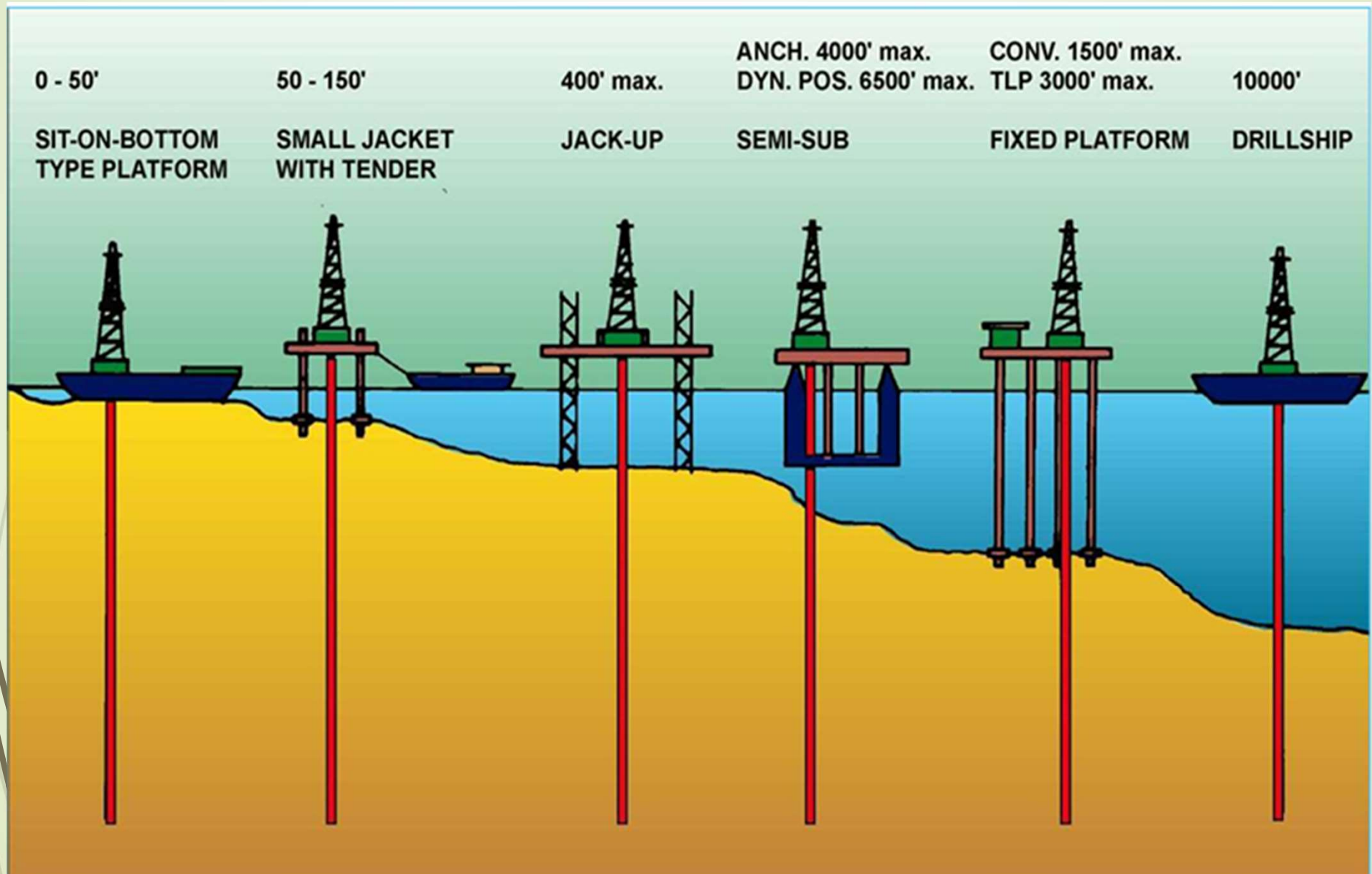
Introduction to Oil & Gas Production

- When offshore oil production first began, only surface wellheads were used / existed.
- Surface means that the “valve” controlling the well pressure is located in a dry environment on a production platform. The “valve” is what is called a Christmas (x-mas) tree.
- With Subsea production systems the x-mas tree is located on the sea bed, on top of the wellhead, and is remotely operated from a nearby platform or a control room onshore.

Surface & Subsea Christmas Tree systems



Offshore Drilling Units: Depth Capabilities



Introduction to Oil & Gas Production

- In terms of production of hydrocarbon, there are 4 types of wells:
 - Oil Production wells.
 - Gas production wells.
 - Water Injection wells.
 - Gas Injection wells.
- Oil is typically processed before it is piped or shipped to refineries.
 - Water / some gas and sand is removed in a processing unit on the platform prior to pumping into a pipeline or offloading it to an oil tanker via a loading buoy.

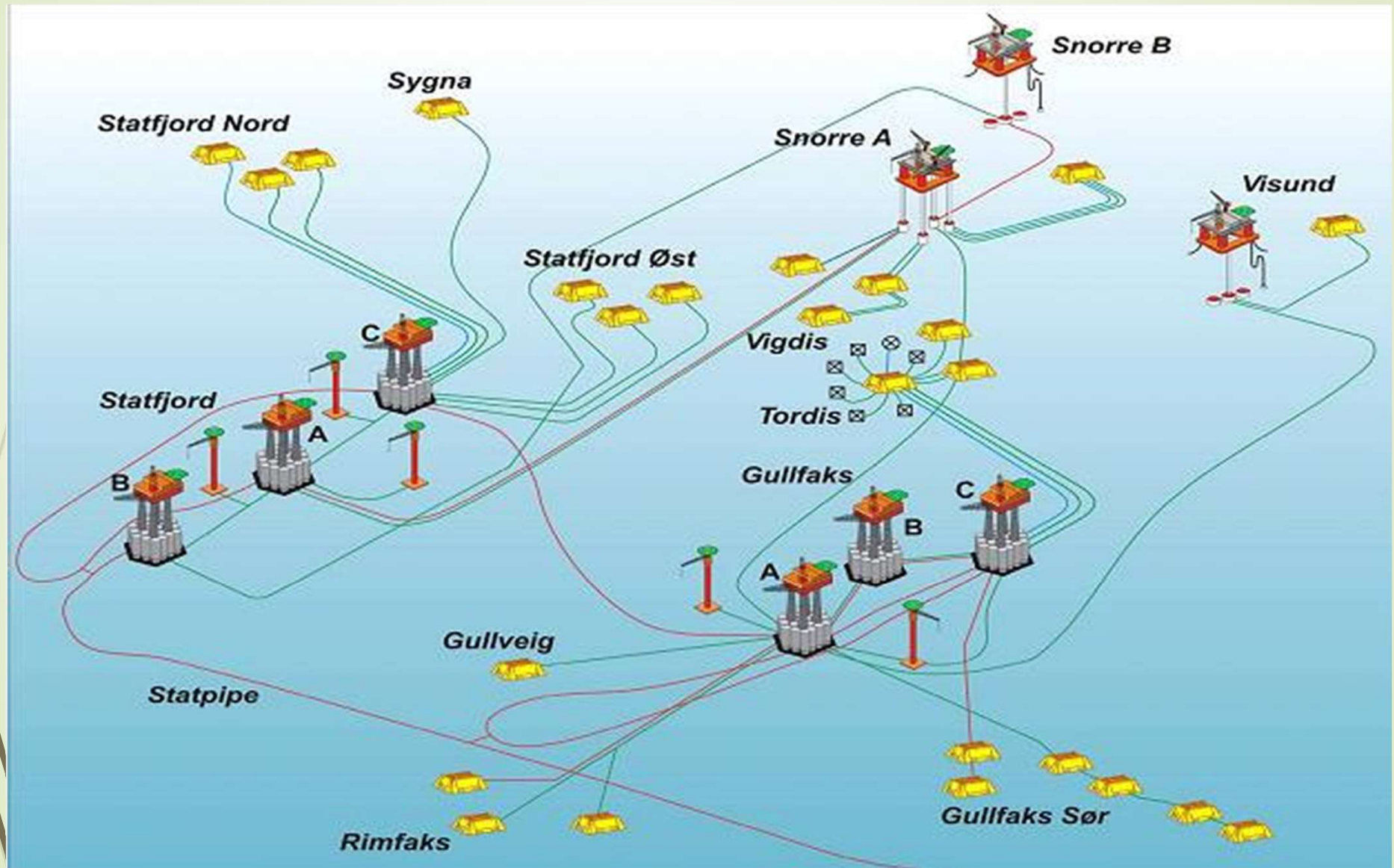
Elements of Subsea Engineering

- There are 4 main elements in Subsea Engineering:
 - Subsea Production System.
 - Flow Assurance and System Engineering.
 - Subsea Structures and Equipment.
 - Subsea Umbilical, Risers and Pipelines.

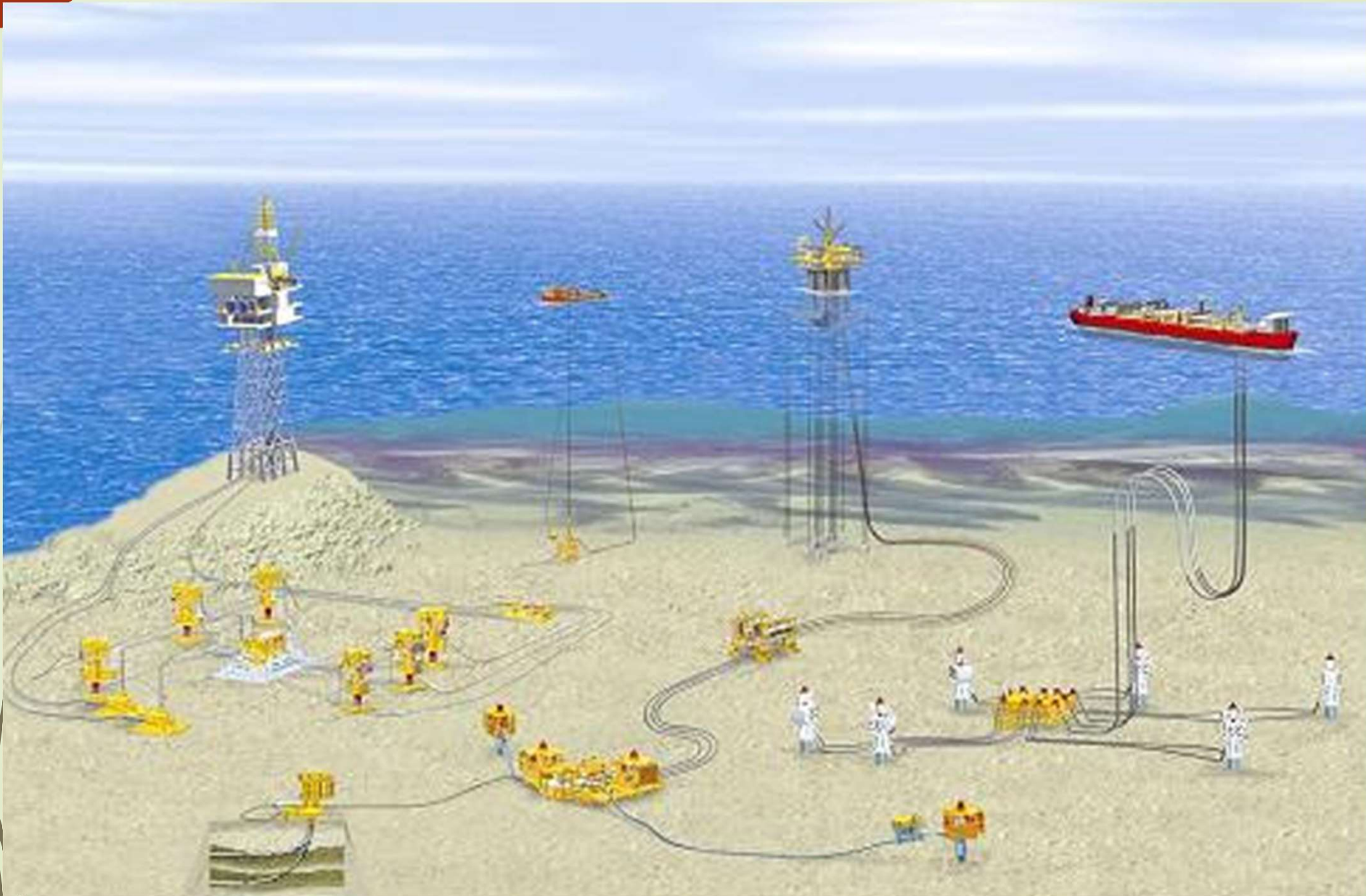
Subsea Production System

- All processes and equipment involved in drilling, field development, and field operation of hydrocarbon resources.
- Subsea production system comprises the following:
 - Deepwater well.
 - Wellhead.
 - Christmas tree.
 - Subsea tie-in to flowline system.
 - Subsea equipment and control facilities to operate the well.

Diagram of a Subsea Production System



Subsea Production System



Subsea Field Development

The following are key considerations in Subsea Field Development:

- Deepwater or shallow-water development.
- Dry tree or wet tree.
- Stand alone or tie-back development.
- Hydraulic and chemical units.
- Subsea processing.
- Artificial lift methods.
- Facility configurations (i.e., template, well cluster, satellite wells, manifolds).

Subsea Distribution System

It involves processes and products that provide communication between subsea controls and topside controls for all equipment via an umbilical system.

- **Topside umbilical termination assembly (TUTA).**
- **Subsea accumulator module (SAM).**
- **Subsea umbilical termination assembly (SUTA), which includes:**
 - Umbilical termination head (UTH).
 - Hydraulic distribution manifold/module (HDM).
 - Electric distribution manifold/module (EDM).
 - Flying leads.
- **Subsea distribution assembly (SDA).**
- **Hydraulic flying leads (HFLs).**
- **Electric flying leads (EFLs).**
- **Multiple quick connector (MQC).**
- **Hydraulic coupler.**
- **Electrical connector.**
- **Logic caps.**

Subsea Surveys

- This involves carrying out a detailed geophysical and geotechnical field development survey together with soil investigation as part of subsea field development.
- This helps in identifying potential Subsea hazards (natural or man-made), and engineering constraints of a proposed subsea field area and pipeline construction.
- It is also used to assess the potential impact on biological communities and to determine the seabed and sub-bottom conditions.

Installation and Vessels

- Subsea installation refers to the installation of subsea equipment and structures in an offshore environment for the subsea production system. Installation in an offshore environment is a dangerous activity, and heavy lifting is avoided as much as possible. This is achieved fully by subsea equipment and structures that are transported to the installation site by installation vessels.
- Therefore, Subsea installation typically requires the use of specialized offshore vessels, which requires diving or robotic equipment.
- Installation of subsea equipment such as trees and templates is typically done by a conventional floating drilling rig, whereas subsea pipelines and subsea risers are installed by an installation barge.

Cost Estimation

- Subsea cost estimation involves developing the cost of the whole subsea project.
- There are 2 cost components:
 - Capital expenditures (CAPEX)
 - Operating expenditures (OPEX).
- CAPEX is the total amount of investment necessary to put a project into operation and includes the cost of initial design, engineering, construction, and installation.
- OPEX is the expenses incurred during the normal operation of a facility, or component after the installation, including labour, materials, utilities, and other related expenses.
- OPEX contains operational costs, maintenance costs, testing costs, and other related costs.

Subsea Control

- Subsea cost estimation involves developing the cost of the whole subsea project.
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- OPEX contains operational costs, maintenance costs, testing costs, and other related costs.

Subsea Power Supply

- Subsea power supply is essential for subsea production system.
- There are 3 aspects of Subsea power supply:
 - Electrical power unit (EPU).
 - Uninterruptible power supply (UPS).
 - Hydraulic power unit (HPU).

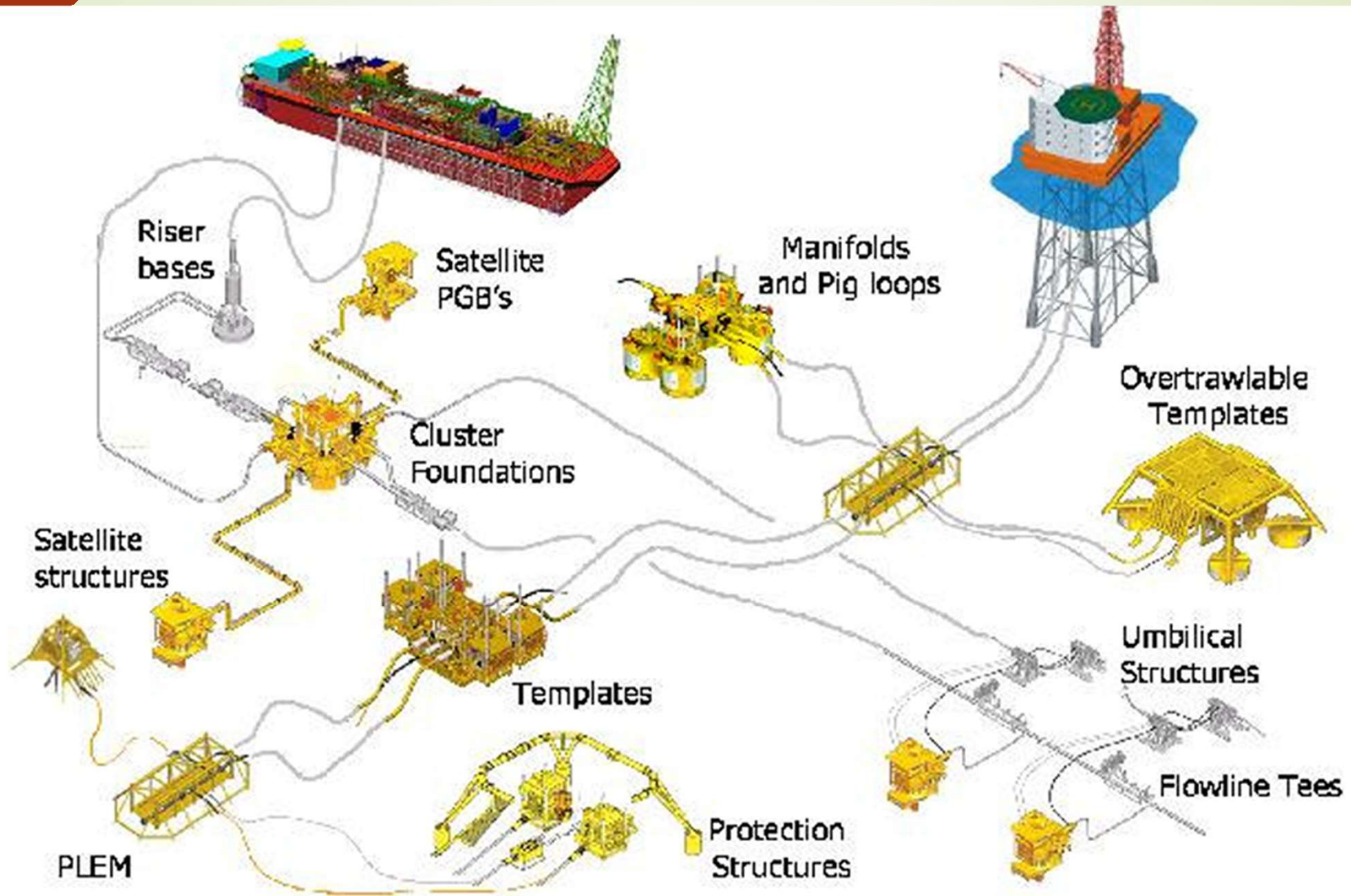
Flow Assurance

- Subsea power supply is essential for subsea production system.
- There are 3 aspects of Subsea power supply:
 - Electrical power unit (EPU).
 - Uninterruptible power supply (UPS).
 - Hydraulic power unit (HPU).

Subsea Operations

- Subsea operations are operations carried out post-installation and they help to ensure safe operations and support the continued flow of hydrocarbons.
- Typical post-installation subsea operations include:
 - Commissioning and start-up (“cold” or “hot” start-up).
 - Normal operations.
 - Production processing.
 - Chemical injection – hydrates, paraffin, asphaltene inhibitors
 - Routine well testing.
 - Inspection, Maintenance and Repairs.
 - Emergency shutdown.
 - Securing facilities (e.g., from extreme weather events).
 - Intervention.

Subsea Structures and Equipment



Subsea Structures and Equipment

Subsea Manifolds.

- Subsea Manifolds are used to gather well streams from, and distribute services to, individual wells.
- A manifold system is used for optimizing as well as controlling the production in a field. This may be done through traditional piping systems, but also through the use of process equipment.
- Manifold piping collects the well stream and enables the production, and distribution, of several wells into designated headers / flowlines. These flowlines may be individual flowlines or bundled flowlines.

Subsea Structures and Equipment

Subsea Manifolds.

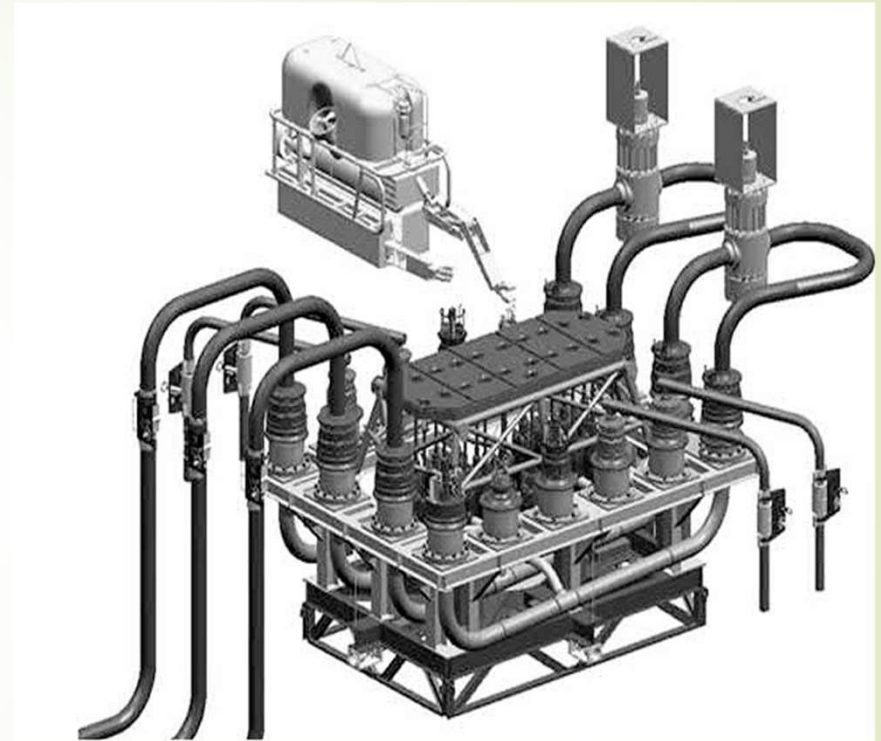
- The numerous types of manifolds range from a simple pipeline end manifold (PLEM/PLET) to large structures such as a subsea process system. The manifold may be anchored to the seabed with piles or skirts that penetrate the mudline. Size is dictated by the number of the wells and throughput, as well as how the subsea wells are integrated into the system.

Subsea Structures and Equipment

Subsea Manifolds.

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Subsea Manifolds



Subsea Structures and Equipment

Pipeline Ends and In-Line Structures.

- The ends of subsea pipeline systems are always connected to a tie-in structure/manifold called pipeline end manifolds (PLEMs).
- A PLEM is a subsea structure set at the end of a pipeline. It is used to connect rigid pipeline with other subsea structures, such as a manifold or tree, through a jumper.
- It is also called a pipeline end termination (PLET), especially to serve as a support for one pipeline valve and one vertical connector.

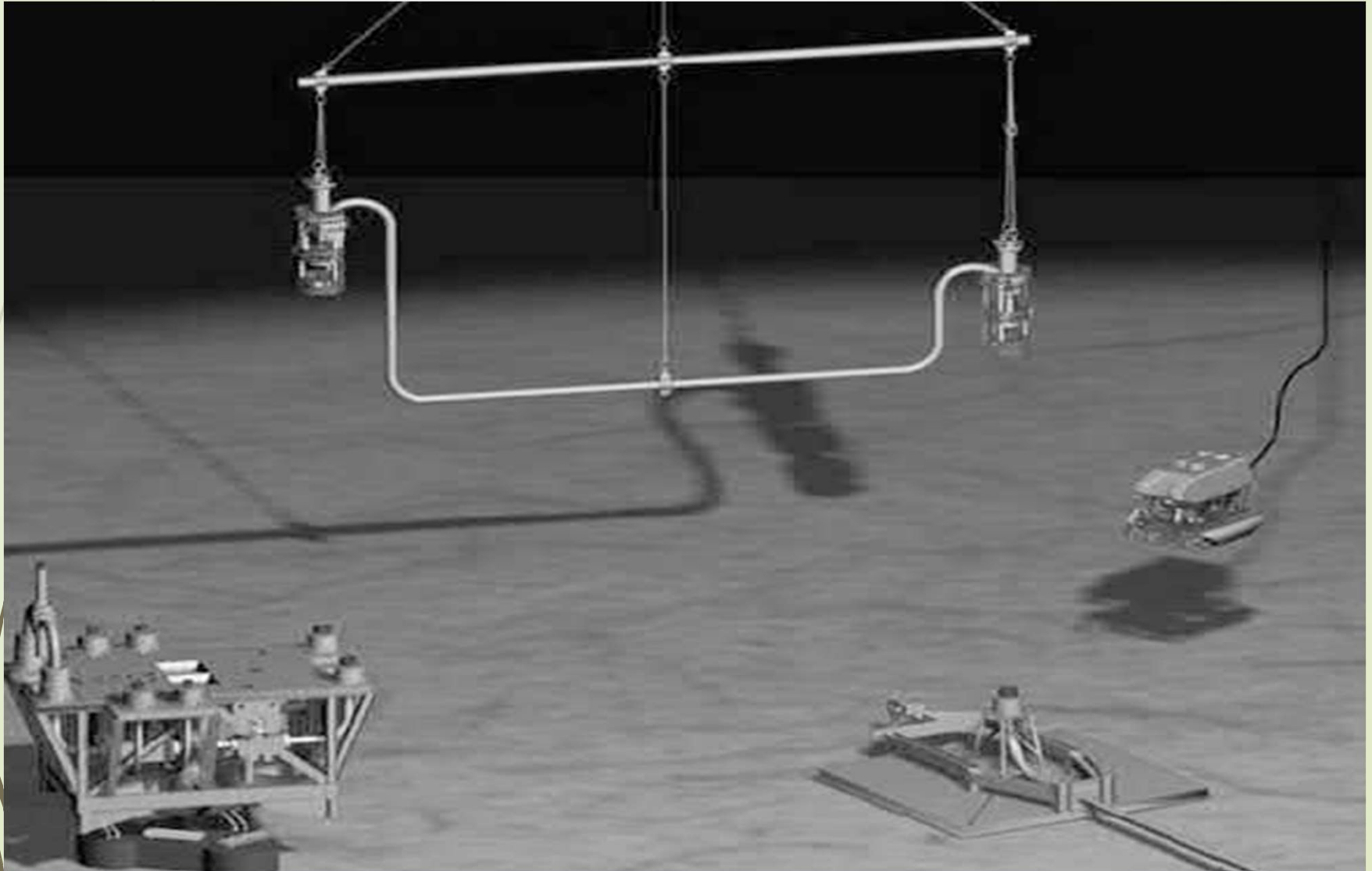
Subsea Structures and Equipment

Jumpers.

- A jumper is a short pipe connector that is used to transport production fluid between two subsea components, e.g., a tree and a manifold, a manifold and another manifold, or a manifold and an export sled.
- It may also connect other subsea structures such as PLEM/PLETs and riser bases. It can also be used to inject water into a well.
- The offset distance between the components (such as trees, flowlines, and manifolds) dictates the jumper length and characteristics.
- Flexible jumper systems provide versatility, unlike rigid jumper systems, which limit space and handling capability.

Subsea Structures and Equipment

Jumpers.



Subsea Structures and Equipment

Subsea Wellheads.

- A Subsea Wellhead is a pressure-containing vessel that serves several functions:
 - It hangs off and seals off casing used in drilling the well.
 - It provides a profile to latch the subsea blowout preventer (BOP) stack and drilling riser back to the floating drilling rig.
 - It acts as the primary pressure barrier for a subsea well.
 - It supports and seals casing strings and also supports the BOP stack during drilling and the subsea tree after completion.

Subsea Structures and Equipment

Subsea Christmas Trees.

- A Christmas tree, or “tree,” is an assembly of a multi-valved structure consisting of valves, pipe spools, and fittings used to control the flow of oil from the well.
- A Christmas tree is the heart of the offshore hydrocarbon production system. It is the primary means of well control and plays a key role in the emergency shutdown system.
- The Christmas tree sits on the top of the wellhead casing system and represents the interface between the well and the production and process facility.

Subsea Structures and Equipment

Christmas Trees.



Subsea Structures and Equipment

Umbilical

- A subsea umbilical is an assembly of hydraulic hoses which can also include electrical cables or optic fibres, used to control subsea structures from an offshore platform or a floating vessel.
- It is an essential part of subsea production system, without which sustained economical subsea petroleum production is not possible.
- The main functions include:
 - Provide hydraulic power to subsea control systems, such as to open/close valves.
 - Provide electric power and control signals to subsea control systems.
 - Deliver production chemicals for subsea injection at tree or downhole.
 - Deliver gas for gas lift operation.

Subsea Structures and Equipment

Risers

- Subsea production Risers are critical for transporting hydrocarbons from the seafloor to the surface.
- They are made up of conductor pipes connected to offshore production platforms or FPSO and the flowlines/Manifolds/Wellheads on the seabed.
- It is also the primary devices of the floating production system to convey fluids to and from vessels. It is one of the most complex aspects of a deepwater production system.
- Subsea Risers are essentially in 2 types: Rigid risers and Flexible risers.
- A hybrid riser is achieved through a combination of these two types of risers.

Subsea Structures and Equipment

Pipelines/Flowlines

- They are used in connecting subsea wellheads to the manifolds or surface facility.
- They are used to transport production, well testing, water injection, gas lift or gas injection. They are either flexible pipes or rigid pipes.
- Some of the requirements for Subsea flowlines include HPHT, sour service flowlines, corrosion resistance



End



Day 2

Course Outline

■ **Overview of Subsea Engineering.**

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Course Objective:

By the end of this course, students will:

Learn and understand Subsea Developments, Equipment, and Installations.



Subsea Field Development

Overview

- Subsea Field Development involves all activities required to produce hydrocarbon resources from a subsea field.
- Water depth is the key parameter that determines the nature of subsea field development.
- A field is called Shallow-water subsea development if the location is less than 200 m (656 ft). In practice, shallow water is the water depth within a diver's reach.
- For a deepwater subsea development, the water depth is between 200 and 1500 m (656 and 5000 ft).
- Ultra-deepwater subsea developments are those in which the water depths are greater than 1500 m (5000 ft).

Wet Trees Vs. Dry Trees

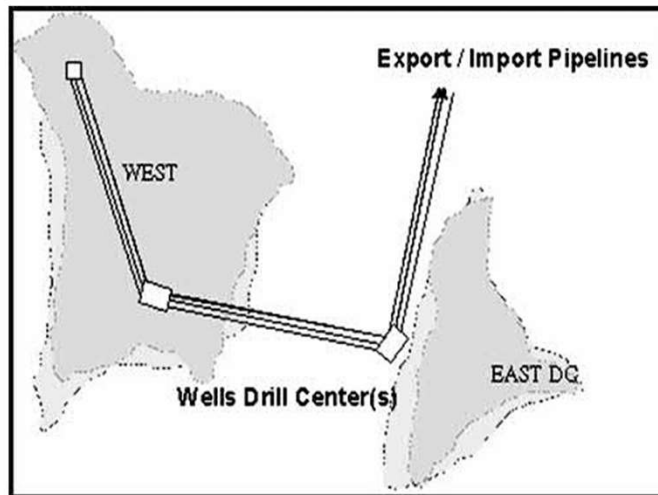
- Dry Trees are usually located on or close to the platform, whereas wet trees can be anywhere in a field in terms of cluster, template, or tie-back methods.
- Dry tree platforms have a central well bay for the surface trees, providing direct access to the wells for workover and recovery.
- Tension leg platforms (TLPs) and spars are normally utilized in a dry tree system.

Wet Trees Vs. Dry Trees

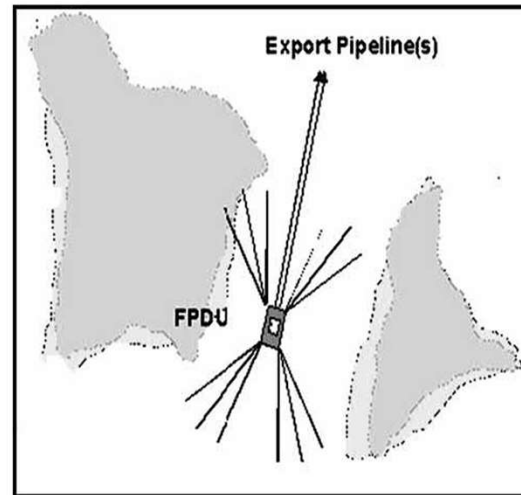
- The subsea field layout for a Wet tree system comes in two types: subsea wells clusters and direct access wells.
- Direct access is mostly applicable in marginal field development.
- All such developments are usually based on semi-submersible floating production and drilling units (FPDUs) with oil export either via pipeline or to a nearby floating storage and offloading (FSO) unit.
- They provide direct and cost-effective access from the surface to the wells to allow for workover or drilling activities directly from the production support, especially for deepwater interventions.

Wet Trees Vs. Dry Trees

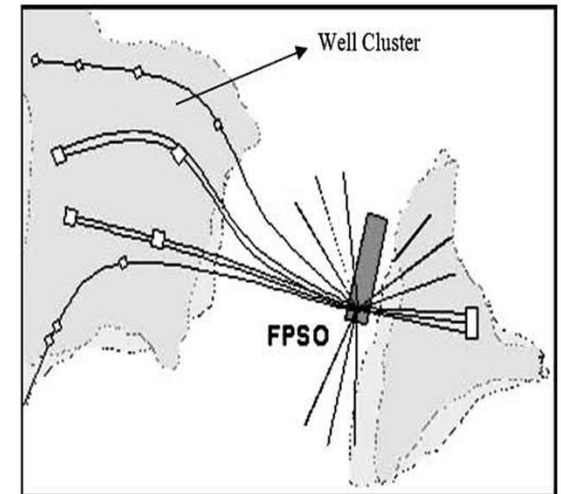
A subsea cluster of wells gathers the production in the most efficient and cost effective way from nearby subsea wells, or (when possible) from a remote /distant sub sea tie-back to an already existing infrastructure based on either a floating production, storage and offloading (FPSO) or a floating production unit (FPU), depending on the region considered.



Tie-Back Field Architecture [1].



FPU Field Architecture [1].



FPSO Field Architecture [1].

Selecting Type of Tree

- To make a decision on which type of Tree (wet tree or dry tree) to install some important parameters are always put into consideration.
- A thorough and objective assessment must be carried out during the selection process and this must include costs, risks, and flexibility considerations to ensure that the development concept that best matches the reservoir characteristics is selected.

Selecting Type of Tree

- **The following are the key criteria for tree selection:**
- **Economic factors:** Estimated NPV, internal rate of return (IRR), project cash flow, project schedule, will most certainly be the key drivers of this choice.
- **Technical factors:** These factors are driven primarily by reservoir depletion plans and means, field worldwide location, operating philosophy, concept maturity and reliability, feasibility, and industry readiness.
- **External factors:** These factors are in the form of project risks, project management, innovative thinking, operator preferences, and people (the evaluation method may vary between each individual).

Subsea Tie-back Development

- Reservoirs with significant hydrocarbon volumes are better placed for subsea development, because the CAPEX and OPEX costs are high and it is difficult to justify the return versus the risk. In some cases, the small marginal oil fields are normally ignored.
- However, in recent times, subsea tie-backs have become popular in the development of marginal oil and gas reserves in an effective and economical way.
- Capital expenditure can be decreased by utilizing the processing capacity on existing platform infrastructures, rather than by continuing to build new structures for every field.
- Subsea tie-backs require significantly lower initial investments, compared with developments using FPSOs or other fixed installations.

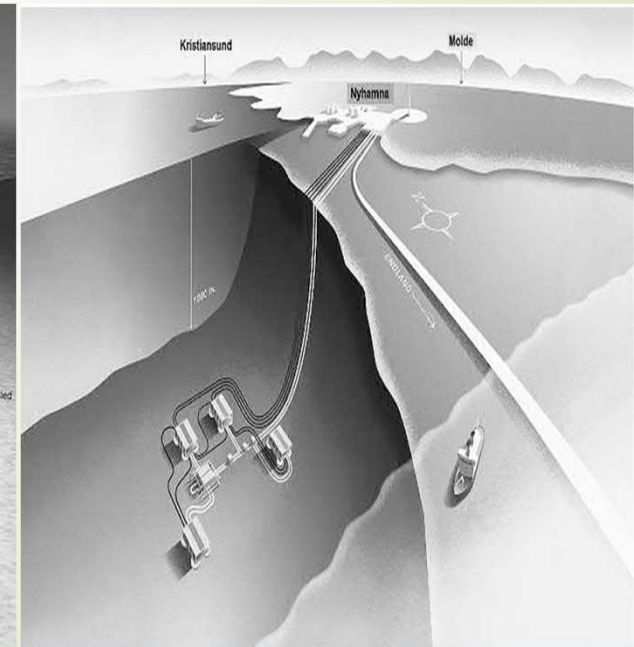
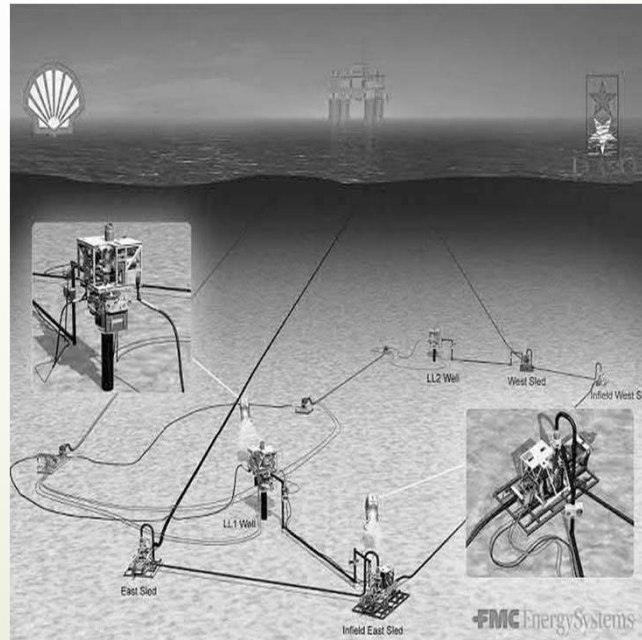
Subsea Tie-back Development

■ The following factors influence the economics of a subsea tie-back system:

- Distance from existing installation.
- Water depth.
- Recoverable volumes, reservoir size, and complexity.
- Tariffs for processing the produced fluids on an existing installation.
- The potentially lower recovery rates from subsea tie-backs versus stand-alone development, due to limitations in the receiving facility's processing systems.
- The potentially higher recovery rates from platform wells, due to easier access to well intervention and workovers.

Subsea Tie-back Development

- There are basically 3 types of Tie-back system:
 - Tie-back to a floating production unit.
 - Tie-back to fixed platform.
 - Tie-back to onshore facility.



Subsea Tie-back Development

- A subsea tie back system includes a subsea wellhead and a flowline to an existing production platform for example.
- Some serious limitations of flow assurance are expected with a longer subsea tie-back, such as hydrate formation induced plugging of the flowline due to the heat loss to the environment and therefore a decrease of temperature along the flowline.
- The conventional remedial methods include thermal insulation of flowlines or injection of chemical inhibitors to prevent the formation of hydrates.
- Such chemicals can be transported from the host platform to the subsea wellhead with an umbilical, and can be injected into the flowline at the wellhead. The umbilical can also be used to control the subsea wellhead. The cost of such umbilical is typically very high.

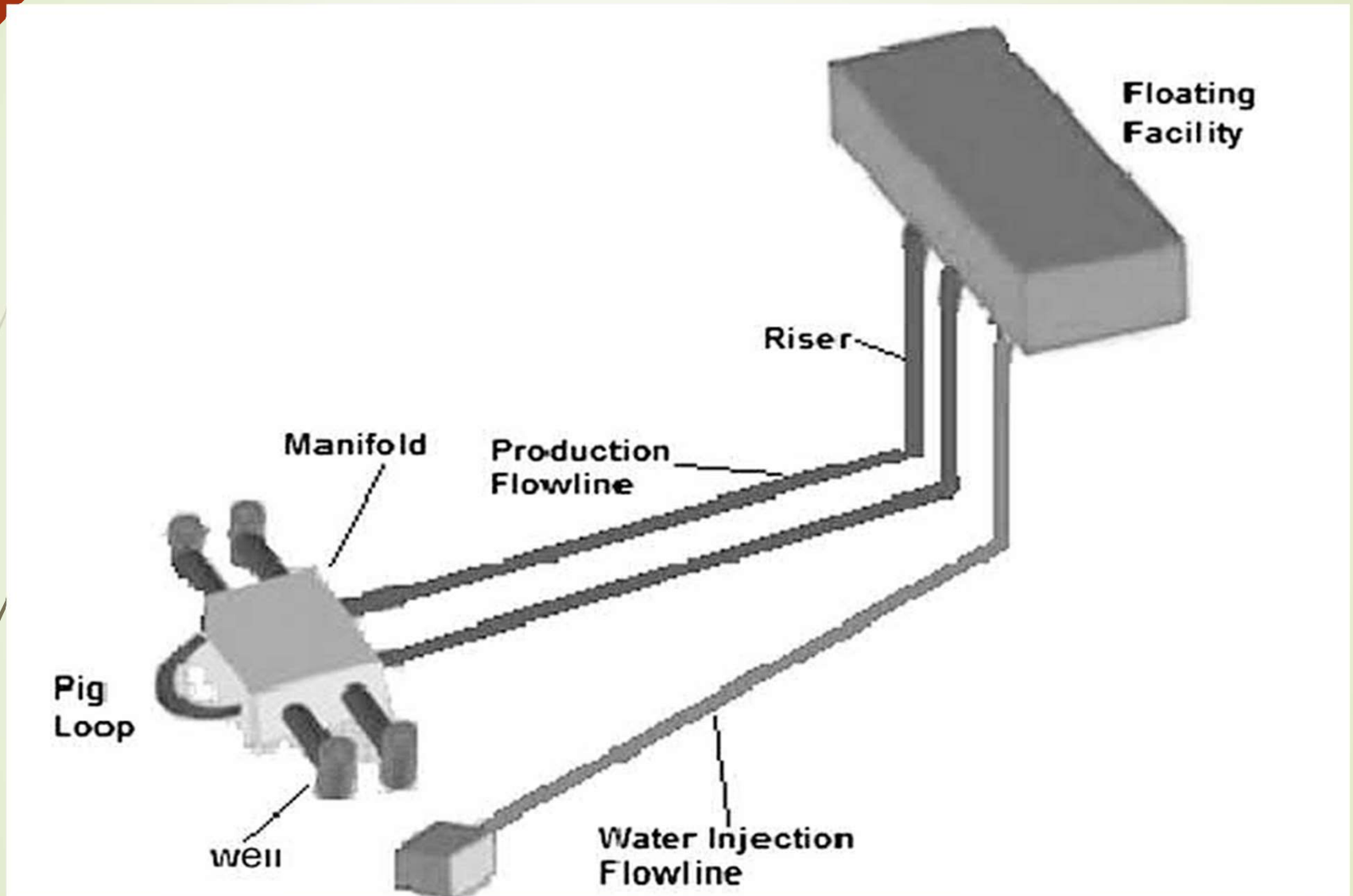
Subsea Tie-back Development

- An alternative development scenario of flow assurance consists of providing a small offshore platform near the wellhead with remote control from the host platform and injection of chemical stored on the small offshore platform via a short umbilical connected to the subsea system.
- The presence of a platform directly above the wellhead enables pig launcher capabilities and well logging tools support.
- When multiphase hydrocarbon flow is expected in the flowline, the tie-back distance is limited because of flow assurance problems.

Subsea Tie-back Development

- Current technological developments are aimed at providing subsea separation facilities on seafloor to separate the fluid into hydrocarbon liquid, gas and water, allowing hydrocarbons to flow over a longer distance.
- Subsea equipment such as subsea pumps may be required to increase the pressure of fluid and assist flow assurance over the tie-back length. Such pump system also requires power which can be provided by a surface facility.
- Dual-flowlines are used widely in the subsea tie-back system to provide a circuit for pig to start from the production platform, go through the flowlines to remove the solids in the flowlines, such as wax, asphaltene, sand, and return to the production platform.

Subsea Tie-back Development



Typical Tie-Back Connections

Tie-back Selection and Challenges

- Advances in flow assurance and multiphase transport now allow the use of tie-backs over much longer distances, while the introduction of subsea processing will strengthen the business case for subsea tie-backs in future field developments.
- Many marginal fields are developed with subsea completions and with the subsea tie-back flowlines to existing production facilities some distance away. Subsea tie backs are an ideal way to make use of existing infrastructure.

Tie-back Selection and Challenges

- **The following are factors in the selection of tie-back system:**
- Cost: Lowest life-cycle cost (i.e. lower CAPEX and OPEX).
- Safety: Safety of personnel and other stakeholders in construction and operation.
- Environment: Impact of development on the environment.
- Technology innovation or transfer: Trial of new technology or transfer of existing technology and know-how.
- Capacity utilization: Use of existing infrastructure, facilities, and elongation of useful life.
- Recoverable volumes, reservoir size, and complexity.
- Tariffs for processing the produced fluids on an existing installation.
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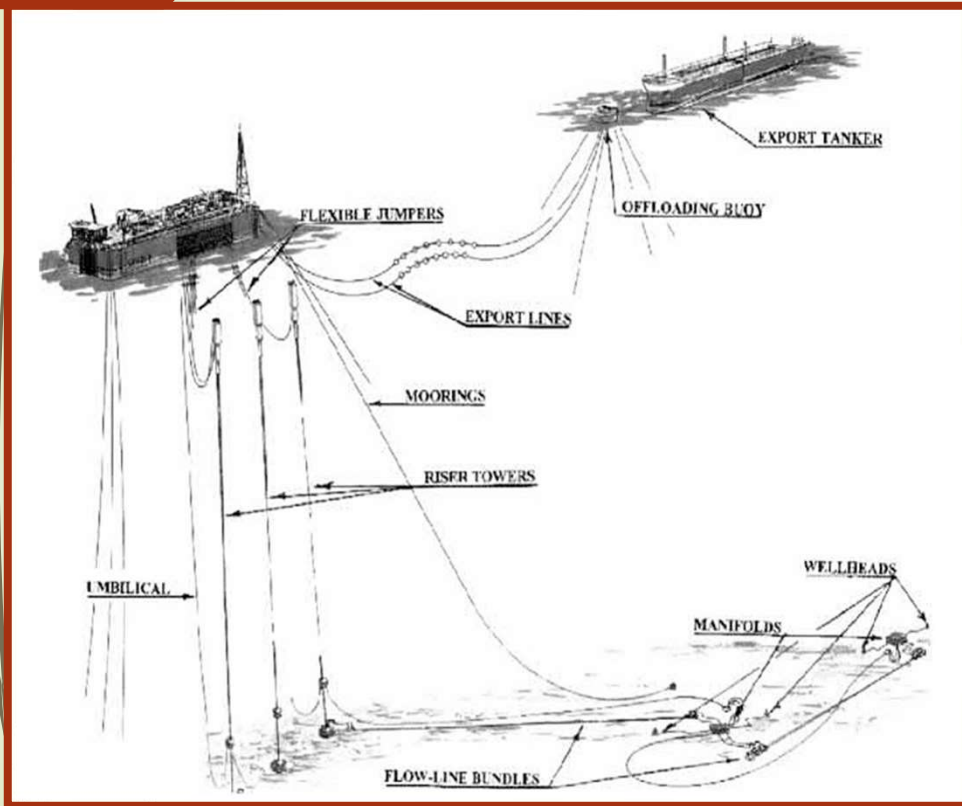
Tie-back Selection and Challenges

- **Long tie-back distances impose limitations and technical considerations:**
- Reservoir pressure must be sufficient to provide high enough production rate over a long enough period to make the development commercially viable. Gas wells offer more opportunity for long tie-backs than oil wells.
- Because of the long distance travelled, it may be difficult to conserve the heat of the production fluids and they may be expected to approach ambient seabed temperatures. Flow assurance issues of hydrate formation, asphaltene formation, paraffin formation, and high viscosity must be addressed. Insulating the flowline and tree might not be enough.
- The gel strength of the production fluids might be too great to be overcome by the natural pressure of the well after a prolonged shutdown. It may be necessary to make provisions to circulate out the well fluids in the pipeline upon shutdown, or to push them back down the well with a high-pressure pump on the production platform, using water or diesel fuel to displace the production fluids.

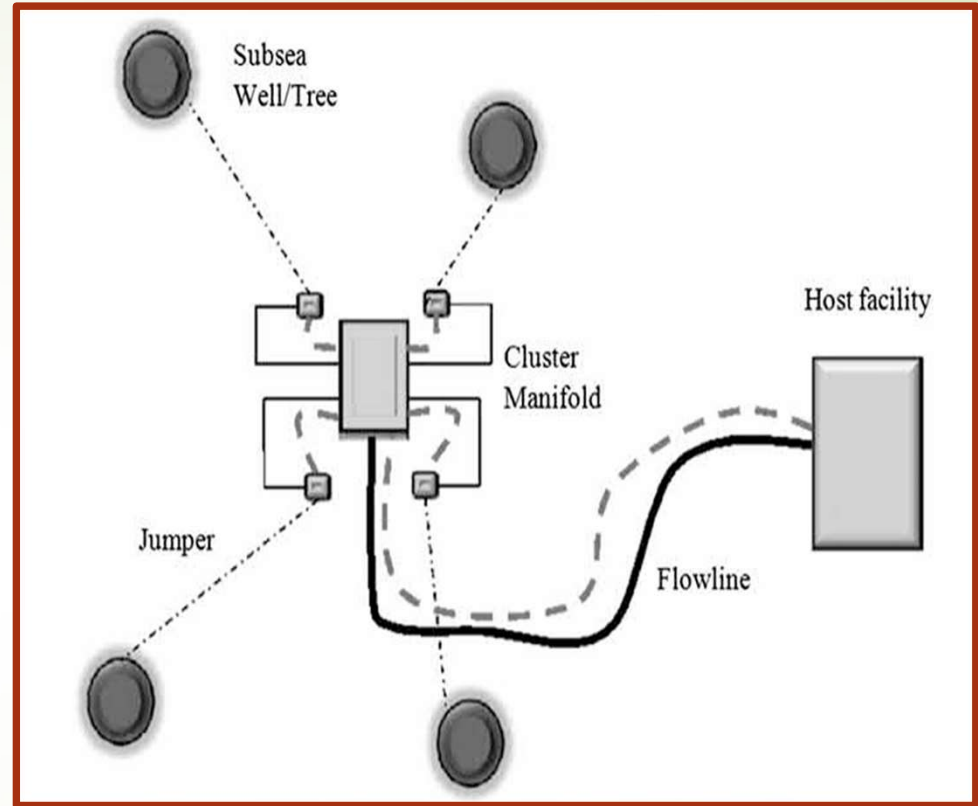
Stand Alone Development

- Stand-alone field development (FD) requires new hardware installation, which is quite expensive in a deepwater environment.
- Making use of the existing infrastructure is the first choice consideration for starting a new development.
- The following issues influence standalone subsea Field Dev.
 - Well groupings.
 - Clustering wells or installing well templates.
 - Optimizing flowline configuration.
 - Pigging requirements.
 - Option of subsea production boosting or pumping as part of the initial development or future needs.

Stand Alone Development



Typical Standalone Field Development Concept



Typical cluster wells concept

Stand-alone Development: Well Grouping

- Well grouping is influenced by the reservoir data and drilling engineering. Well types and locations can be determined once the reservoir is mapped and the number of wells is created according to the reservoir model.
- Well grouping categories:
 - Satellite wells: typically used for small field development requiring few wells, for example, concept of tie-back to the existing structures.
 - Cluster wells: common concept for a stand-alone FD. Typically there are 3 to 8 subsea XTs in a manifold.
 - Template wells: wells are grouped closely together. This concept usually utilizes a template in which the well guide bases and the manifold are integrated. Subsea Xmas trees are landed/locked on each slots of the template.
 - Combination of the above.

Differences between Stand-alone and Tie-back Subsea Field Development

Concept	Features
Subsea tie-back development	• Investment is reduced using the spare capacity in the existing platform; • Very suitable option for small developments e.g. marginal fields;
Stand-alone development	• No existing subsea structures and the recoverable reserves are large; • Or the distance to shore is too far; • New stand-alone development can be considered using a platform that is the most suitable for the environmental conditions.

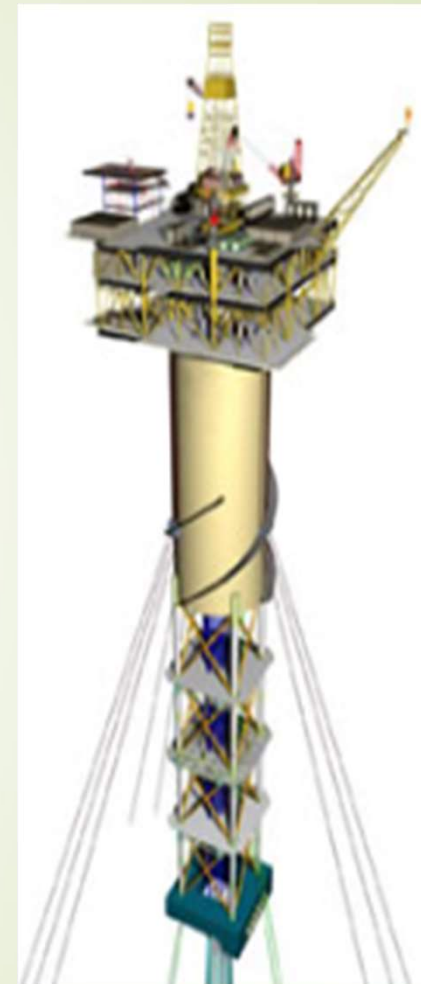
Facilities used for Stand-alone FD.

- Facilities used for Standalone FD are divided into 2 categories: fixed platforms and floating systems.
- Types of Facilities:
- Fixed platforms: They are built on concrete or steel jackets which are directly anchored on the seabed. Various types of platforms, e.g. concrete caisson & steel jacket.
 - Fixed platforms are usually installed in the water depth within 500m.
- TLP: tension leg platform (TLP) is a vertically moored floating structure. A group of tethers called “leg” is used to moor the each corners of the floating structure. Lateral movement is allowed, but vertical movement is prevented by the legs. The TLP can be used in the water depth from 300 m to 1500m.
 - Both dry trees and wet trees can be used for the TLP production system.

Facilities used for Stand-alone FD.

- Facilities used for Standalone FD are divided into 2 categories: fixed platforms and floating systems.
- Types of Subsea FD Facilities:
- SPAR: the Spar platform utilize a large-diameter, single vertical cylinder buoy floating on the sea water surface to support the decks. Can be used in water depth of 1500 m.
- Semi-submersible platforms: They can be operated for drilling, subsea equipment installation and oil/gas production.
- FPSO: a floating vessel for the processing and storage of oil and gas. It is designed to receive the production oil and gas from nearby platform or subsea production systems. The oil is offloaded onto a tanker or transported through pipelines.

Facilities used for Stand-alone FD.



A SPAR

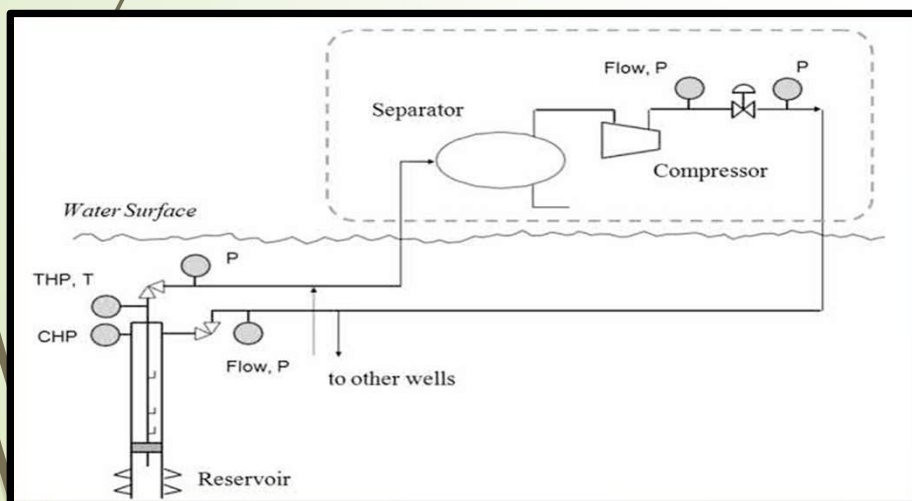
Host Facilities for Subsea Field Development

Artificial Lift Methods

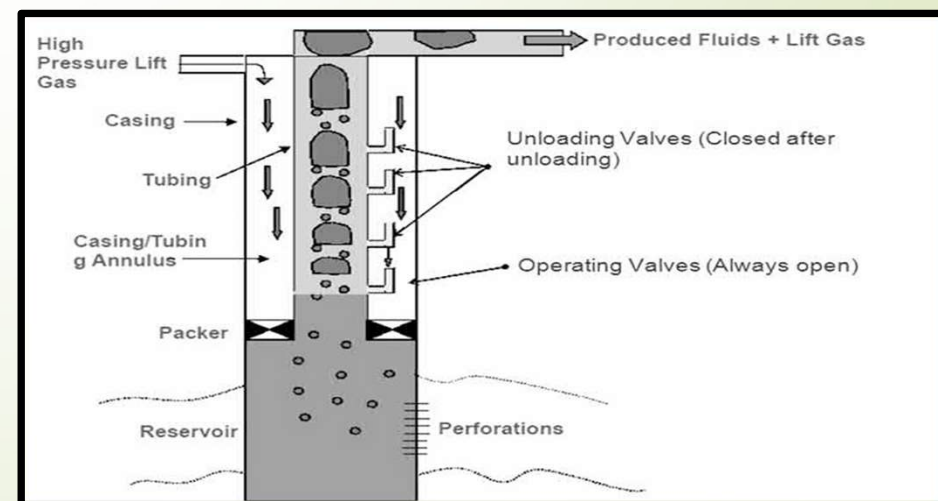
- Artificial lift is cheaper in land/shallow water operations.
- However, most deepwater oil fields ultimately require artificial lift to maintain production flow and achieve economic objectives.
- Upfront planning for an artificial lift in deepwater is critical, as the environment is operationally more difficult and economically more challenging.
- The following are the most popular artificial lift methods:
 - Gas Lift (GL).
 - Subsea Boosting.
 - Electrical submersible pumping (ESP).

Gas Lift Installation

- Gas lift (GL) is a predominant artificial lift method used in the offshore environment to date.
- However, GL applications become more limited due to higher operating pressures and ESPs applications become more suitable.
- The following factors will lead to Gas lift installation:
 - High water cut of the reservoir.
 - Low GOR of the reservoir.
 - Long offset flowline.



Configuration of Typical Subsea GL System



Details of Gas Lift in Downhole

Gas Lift Installation

- Design of Gas lift system involves 2 key parameters:
 - Gas lift volume.
 - Gas lift pressure.
- Gas lift volume is the total requirement for the field determined by individual well requirements.
- Production will increase as a function of lift gas volume until a point of maximum production is reached.
- Gas lift pressure should be determined carefully since this parameter will influence the system operating pressure, material and equipment specification of the well system.

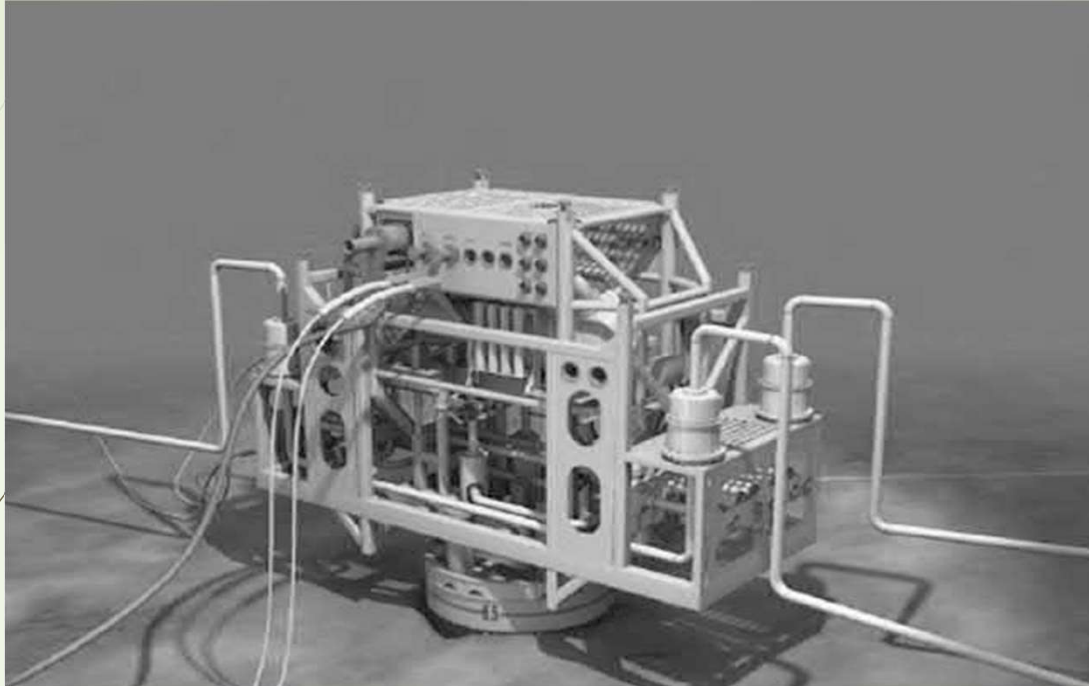
Subsea Pressure Boosting

- Some reservoirs have sufficient pressure to push the production fluid from the reservoir to the surface without the use of downhole artificial lift.
- However, due to the reduction of the reservoir pressure after a long time production, or because of ultra-deepwater light-oil and deepwater heavy-oil reservoirs having pressure that is near hydrostatic pressure, it is quite difficult to produce the fluids to the sea surface.
- Subsea boosting as shown in Figure 2-16 reduces or eliminates the backpressure on the wells resulting from the riser hydrostatic head and the riser and flowline pressure drop caused by high viscosity. The pressure increase between the output of the boosting and the backpressure on the well will increase the flow from the well. Components of a subsea boosting station may include:

Subsea Pressure Boosting

- Subsea booster pumps reduces the backpressure on the wells resulting from the riser hydrostatic head and the riser and flowline pressure drop caused by high viscosity.
- Components of a subsea boosting station may include:
 - Subsea gas compressor.
 - Subsea multiphase pump.
 - Subsea wet gas compressors.
- Subsea pressure boosting system enables longer subsea tiebacks, which potentially could enable the economics of exploiting small, remote, marginal fields.
- Subsea separation provides an economic alternative existing surface facilities, allowing better utilization of these installations by adding new sub sea tiebacks which currently would not be economic to develop.
- Subsea separation allows separation at the seabed and transported of fluid streams to different production facilities, solving flow assurance problem due to low temp at seabed

Subsea Pressure Boosting



Pressure booster pump

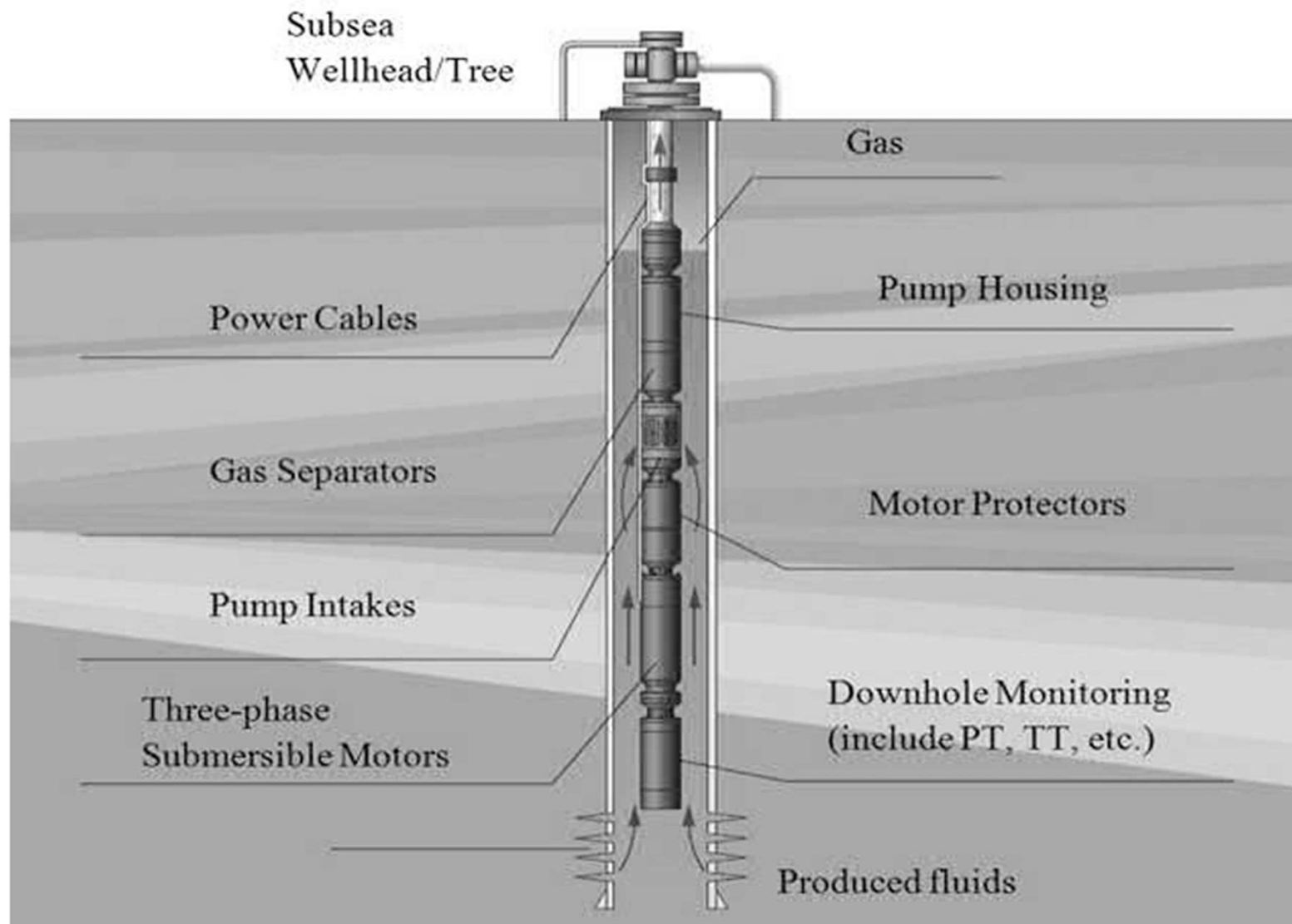
Electric Submersible Pumps (ESP)

- ESP installation is an effective and economical method of lifting large volume of reservoir fluids from downhole under different well conditions.
- They require a large electricity supply, but it is less complex and more efficient than delivering gas to gas lift systems.
- Unlike the surface pump system, ESP systems are designed to be immersed in the fluid, located either in a well or on the seabed.
- The ESP motors are pressure balanced with the environment, either downhole pressure or water pressure in subsea conditions.

Electric Submersible Pumps (ESP)

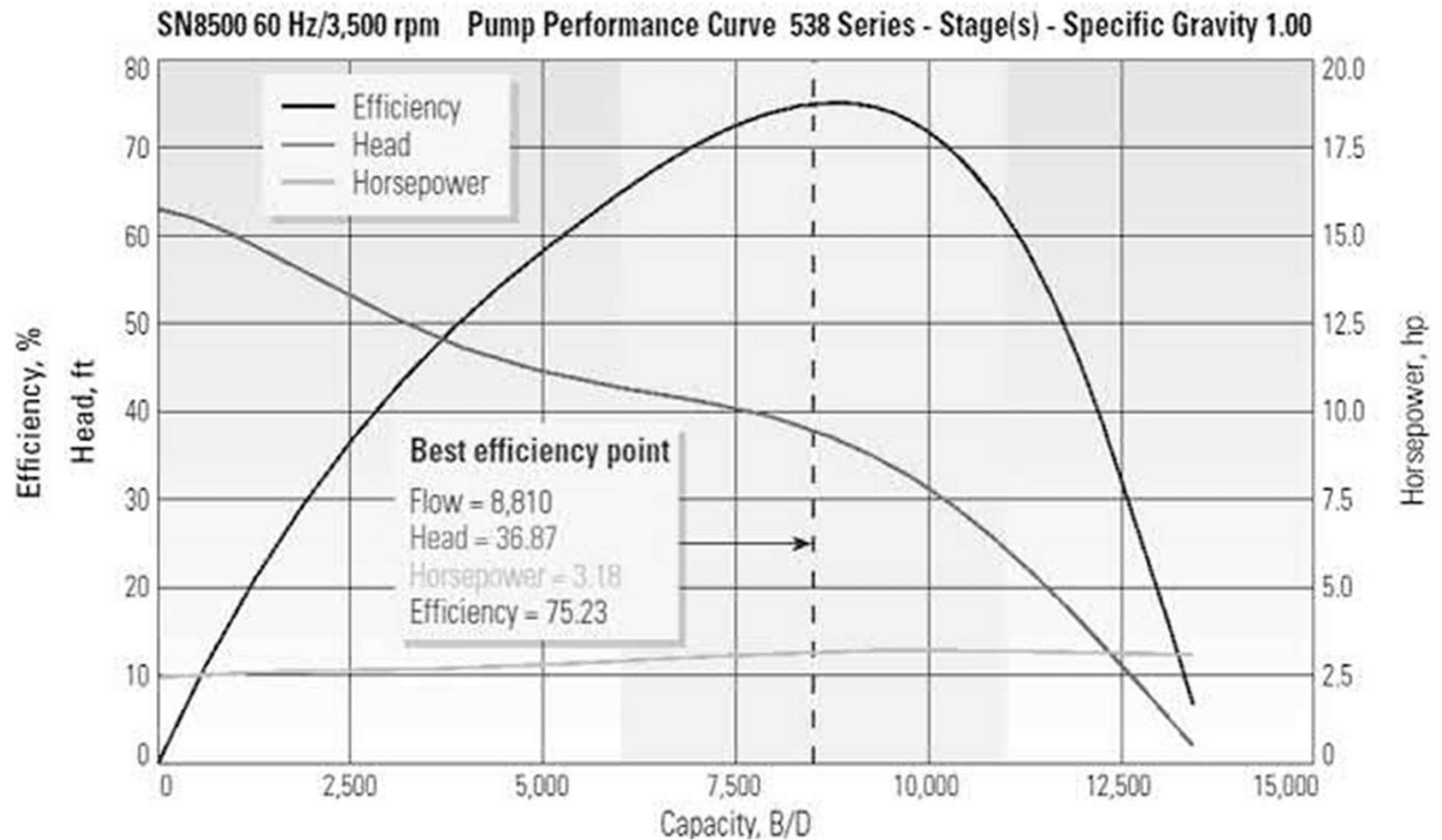
- Sizing of ESP is based on predicted completion performance, or flow rate.
- This usually involves examination of the well inflow performance relationship (IPR), which describes the production response to changes in bottomhole pressure (BHP).
- Data required for calculation and sizing of ESP includes:
 - well data
 - production data
 - well fluid conditions
 - power sources
 - possible problems etc.

Electric Submersible Pumps (ESP)



Typical Downhole ESP Configuration

Electric Submersible Pumps (ESP)



Pump Performance Curve

Subsea Processing

- Subsea processing (SSP) is the process of handling and treatment of the produced fluids for mitigating flow assurance issues prior to reaching the platform or onshore.
- This involves the following:
 - Pressure Boosting.
 - Separation.
 - Solids management.
 - Heat exchanging.
 - Gas treatment.
 - Chemical injection.
- Subsea separation can be either 2- or 3-phase separation:
- Two-phase separators are used for separation of any gas-liquid system such as gas-oil, gas-water, and gas-condensate systems.
- Three-phase separators are used to separate the gas from liquid phase and water from oil.

Subsea Processing

- **Advantages of Subsea processing in a Subsea Field Dev. are:**
 - **Reduced total CAPEX -reducing the topside processing and/or pipeline CAPEX.**
 - **Accelerated and/or increased production and/or recovery.**
 - **Enabling marginal field developments, especially fields at deepwater/ultra-deepwater depths and with long tie-backs.**
 - **Extended production from existing fields.**
 - **Enabling tie-in of satellite developments into existing infrastructure by removing fluid.**
 - **Handling constraints.**
 - **Improved flow management.**
 - **Reduced impact on the environment.**

Subsea Field Development Assessment

- Subsea field layout designs are selected based on historical data obtained from other projects, and technical inputs and insights from various subdisciplines within the organisation.
- Some critical design factors include the following:
 - Engineering and design.
 - Cost and schedule involved.
 - Well placement and completion complexity.
 - Flexibility of field expansion.
 - Ease of construction and fabrication for the subsea hardware.
 - Intervention capability (easy, moderate, or difficult to intervene).
 - Rig movement (mild to extreme environmental conditions).
 - Installation and commissioning, such as ease of installation and commissioning, flexibility of installation sequence.
 - Reliability and risk of the field architecture.
 - ROV accessibility.

Process Simulations

- Typical Well Data input into process flow simulations:
- Pressure $\frac{1}{4}$ 8000 psia.
- Temperature $\frac{1}{4}$ 200F.
- Well PI $\frac{1}{4}$ 20 bpd/psi.
- Reservoir pressure maintenance by water injection.
- True vertical depth (TVD) $\frac{1}{4}$ 11,000 ft from mudline.
- Tubing size $\frac{1}{4}$ 5.5-in. OD (0.36-in. WT).
- Kick-off point $\frac{1}{4}$ 2,000 ft below mudline.
- Kick-off angle $\frac{1}{4}$ 45.
- Roughness of tubing $\frac{1}{4}$ 0.0018".
- U value for wellbore $\frac{1}{4}$ 2.0 Btu/(ft² hr F).
- Water depth of 8000 ft.

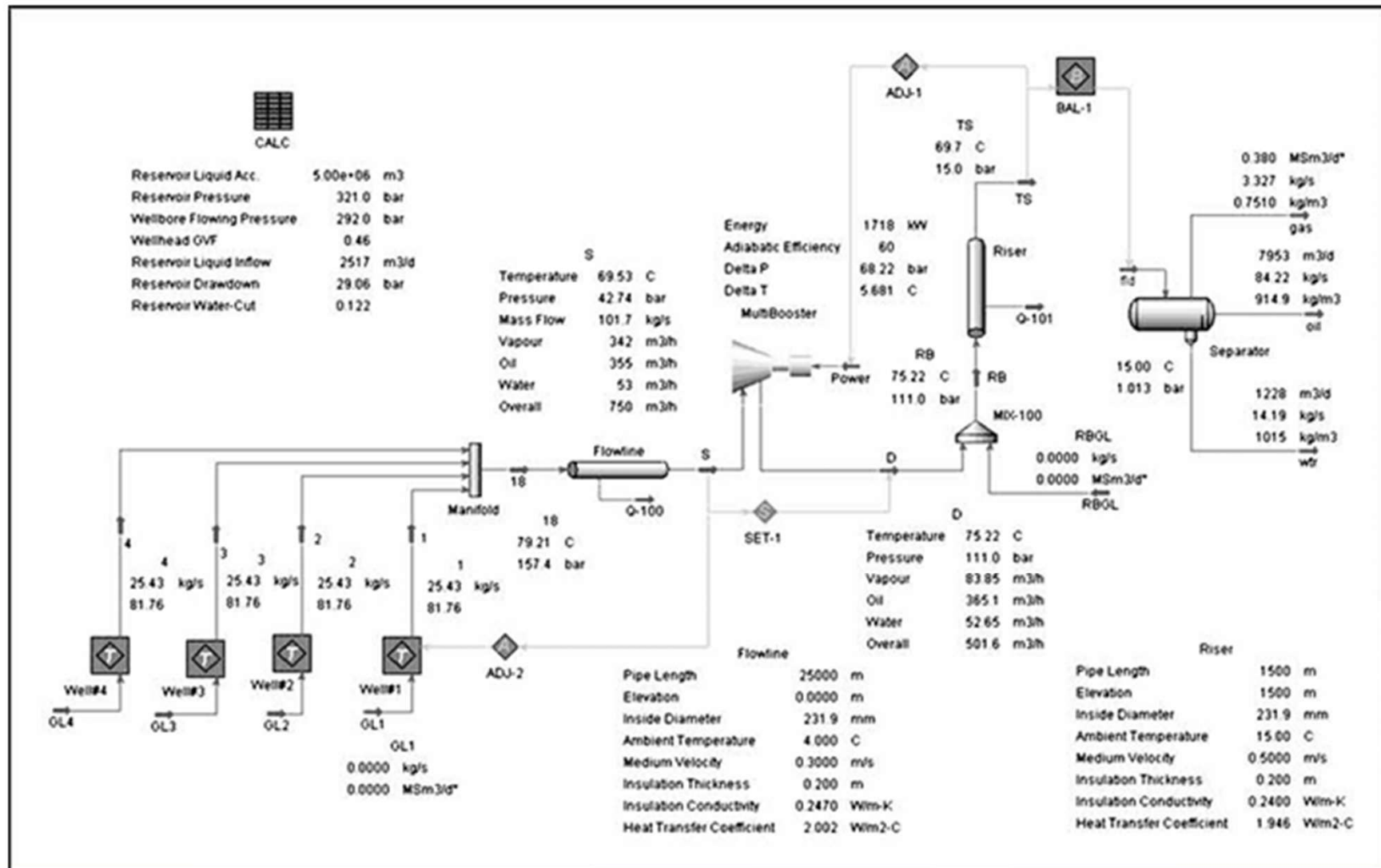
Process Simulations

- Typical Well Data input into process flow simulations (cntd):
- Subsea well completion, dual flowlines, three identical wells per flowline.
- Tie-back distance from subsea manifold to topsides (flowline þ riser) of 25,000 ft.
- Steel catenary riser, pipe-in-pipe configuration considered for both riser and flowline with an assumed U value.
- Seabed terrain assumed to be flat.
- Typical environmental conditions including seawater temperature profile and air temperature were used.
- Riser/flowline roughness $\frac{1}{4}$ 0.0018 in.
- 10- and 12-in. nominal riser/flowline sizes.

Process Simulations

- Process flow simulations are often carried out using HYSYS process simulator.
- Pressure drop calculations in different pipe segments are carried out using OLGAS.
- Typically, a series of simulations are performed for various reservoir accumulations to establish the liquid production from the reservoir as a function of the reservoir condition. These simulations are then repeated for the various production cases that are investigated.
- The time it takes to produce a certain volume from the reservoir is calculated by integrating the liquid rate function with regard to volume accumulation.

Process Simulations



Process Simulation Model